

daily price movement. Furthermore, investors could potentially manipulate the PWP by trading at a more aggressive rate to push the price up for buy orders or down for sell orders, and give the market the impression that they still have more to trade. Since temporary impact does not dissipate instantaneously, the PWP price computed over a slightly longer horizon could remain artificially high (buy orders) or artificially low (sell orders) due to temporary impact cost. Participants may hold prices at these artificially higher or lower levels waiting for the non-existent orders to arrive. The end result is a PWP price that is more advantageous to the investor than what would have occurred in the market if the order had actually traded over that horizon.

Relative Performance Measure (RPM)

The relative performance measure (RPM) is a percentile ranking of trading activity. It provides an indication of the percentage of total activity that the investor outperformed in the market. For a buy order, it represents the percentage of market activity that transacted at a higher price and for a sell order it represents the percentage of market activity that transacted at a lower price. The RPM is modeled after the percentile ranking used in standardized academic tests and provides a descriptive statistic that is more consistent and robust than other measures.

The RPM was originally presented in *Optimal Trading Strategies* (2003) and Kissell (2007) and was based on a volume and trade metric. That original formulation, however, had at times small sample size and large trade percentage limitations bias. For example, the original formulation considered all of the investor's trades at the average transaction price as outperformance. Therefore, in situations where the investor transacted a large size at a single price all the shares were considered as outperformance and the end result would overstate the actual performance. Leslie Boni (2009) further elaborates on this point in her article "Grading Broker Algorithms," *Journal of Trading*, Fall 2009, and provides some important insight and improvements.

To help address these limitations, we revised the RPM formulation as follows:

The RPM is computed based on trading volume as follows:

$$\begin{aligned} \text{RPM} = & \frac{1}{2} \cdot ((\% \text{ of volume traded at a price less favorable or equal to } P_{\text{avg}})) \\ & + (1 - \% \text{ of volume traded at a price less favorable or equal to } P_{\text{avg}})) \end{aligned} \quad (3.17)$$